AN IRREDUCIBLE GROUP SCHEME OVER A FIELD IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-group-scheme-field-geometrically-irreducible (canonical tag 047M). cited at proof lines 1139-1141. Irreducibility persists under field extensions.
- Lemma varieties-lemma-make-Jacobson (canonical tag 0479). cited at proof lines 1143-1148. Reduction to Jacobson with rational closed points.
- Lemma lemma-group-scheme-over-field-open-multiplication (canonical tag 047K). cited at proof lines 1155-1157. Openness of the multiplication image.

Source reference. Original lemma 047M: sources/stacks-chow/tags.txt, source lines 5513–5513. Cited at proof lines 1139-1141 to see G_K stays irreducible; load-bearing for the reduction.

Source reference. Original lemma 0479: sources/stacks-chow/tags.txt, source lines 5500–5500. Cited at proof lines 1143-1148 to reduce to G Jacobson with k-rational closed points; load-bearing.

Source reference. Original lemma 047K: sources/stacks-chow/tags.txt, source lines 5511–5511. Cited at proof lines 1155-1157 to see the image is open; load-bearing.

Lemma 0.2. *Let G be a group scheme over a field k. If G is irreducible, then G is quasi-compact.*

Proof. Suppose that K/k is a field extension. If G_K is quasi-compact, then G is too as $G_K \to G$ is surjective. By Lemma 047M we see that G_K is irreducible. Hence it suffices to prove the lemma after replacing k by some extension. Choose K to be an algebraically closed field extension of very large cardinality. Then by Varieties, Lemma 0479, we see that G_K is a Jacobson scheme all of whose closed points have residue field equal to K. In other words we may assume G is a Jacobson scheme all of whose closed points have residue field k.

Let $U \subset G$ be a nonempty affine open. Let $g \in G(k)$. Then $gU \cap U \neq \emptyset$. Hence we see that g is in the image of the morphism

$$U \times_{\mathrm{Spec}(k)} U \longrightarrow G, \quad (u_1, u_2) \longmapsto u_1 u_2^{-1}$$

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Since the image of this morphism is open (Lemma 047K) we see that the image is all of G (because G is Jacobson and closed points are k-rational). Since U is affine, so is $U \times_{\mathrm{Spec}(k)} U$. Hence G is the image of a quasi-compact scheme, hence quasi-compact. $\square$